

THE LYRE

Lyra



SIZE RANKING 52

BRIGHTEST STAR

Vega (α) 0.0

GENITIVE Lyrae

ABBREVIATION Lyr

HIGHEST IN SKY AT 10 PM
July–AugustFULLY VISIBLE
90°N–42°S

Lyra lies on the edge of the Milky Way next to Cygnus and is a compact constellation of the northern sky. It includes Vega, or Alpha (α) Lyrae (see p.253), which is the fifth-brightest star in the sky and one of the so-called Summer Triangle of stars—the other two being Deneb (in Cygnus) and Altair (in Aquila). The Lyrid meteors radiate from a point near Vega around April 21–22 every year. Lyra represents the stringed instrument played by Orpheus (see panel, below).

SPECIFIC FEATURES

Vega dazzles at magnitude 0.0, appearing somewhat blue-white in color to the unaided eye. It is

the standard star against which astronomers compare the color and brightness of all other stars.

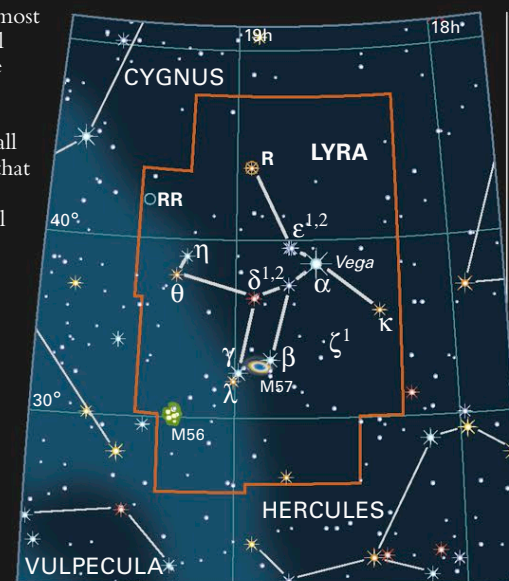
The finest quadruple star in the sky—Epsilon (ϵ) Lyrae (see p.276)—is found three diameters of a full moon from Vega. Binoculars easily show it as a neat pair of 5th-magnitude white stars, but each of these has a closer companion that is brought into view with a telescope of 2.5–3 in (60–75 mm) aperture and high magnification. All four stars are linked by gravity and are in long-term orbit around each other.

Two other double stars near Vega that are easy to identify with binoculars are Zeta (ζ) and Delta (δ) Lyrae, each with components of 4th and 6th magnitudes. Beta (β) Lyrae is another double star, easily resolved by a small telescope into its cream and blue components. The brighter star (the cream one) is an eclipsing binary that fluctuates between magnitudes 3.3 and 4.4 every 12.9 days. Many years of study have established that Beta's two stars are so close that gas from the larger of the pair falls toward the smaller companion and some of it spirals off into space. Almost midway between Beta

and Gamma (γ) Lyrae lies the most photographed of Lyra's celestial treasures, the Ring Nebula (see p.257), or M57. This planetary nebula is shaped like a smoke ring and appears through a small telescope as a disk larger than that of Jupiter. Larger apertures are needed to make out the central hole. Studies with the Hubble Space Telescope have revealed that the “ring” is in fact a cylinder of gas thrown off from the central star, oriented almost end-on to the Earth.



THE LYRE



THE RING NEBULA ☾

One of the most famous planetary nebulae in the whole sky, the Ring Nebula, or M57, consists of hot gas shed from a central star. Its beautiful colors are revealed only in photographs such as this one.

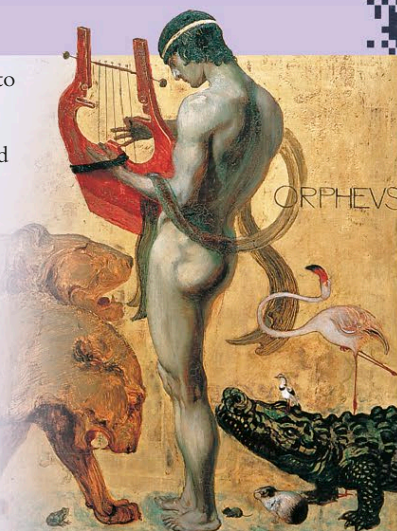
MYTHS AND STORIES

ORPHEUS

Heartbroken Orpheus descended into the Underworld to retrieve his wife, Eurydice, who had been killed by a snake. His songs charmed Hades, god of the Underworld, who agreed to release Eurydice, provided Orpheus did not look back as he led her to the surface. At the last minute, Orpheus glanced behind him, and Eurydice faded away. Orpheus then roamed the Earth, disconsolately playing his lyre.

ENTRANCED

Orpheus was said to have charmed even the rocks and streams with his music. In this 19th-century painting, he tames the wild animals with his songs.



STRINGED INSTRUMENT 👁

Lyra, dominated by dazzling Vega, represents the harp played by Orpheus, the musician of Greek myth. Arab astronomers visualized the constellation as an eagle or vulture.